

Trabajo unidad 2: Hardening de sistemas operativos Kali Linux

Carrera: Ingeniería en ciberseguridad

Asignatura: Hardening y seguridad de sistemas operativos

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1. Introducción

El hardening de los sistemas operativos es un conjunto de herramientas, metodologías y prácticas para acotar el campo en el que puedan actuar los ciberdelincuentes frente a la oportunidad de la generación de nuevas y antiguas vulnerabilidades generadas en el tanto a nivel de hardware, software y sistemas de información de los cibernautas. Por ello, su aplicación en el mundo de la seguridad de datos es fundamental frente a la continua exploración y explotación de nuevas vulnerabilidades por parte de ciberatacantes, promoviendo las buenas prácticas y la seguridad de la información. (S/f. Ninjaone.com)

En el siguiente trabajo se hará un proceso de hardening básico a nivel de sistema operativo Linux, en específico kali Linux. Donde finalmente se podrá evaluar la presencia de vulnerabilidades y si el proceso de hardening fue efectivo o falta aplicar nuevas medidas de seguridad.

Sistema Operativo :Kali Linux

2. Actualización del sistema

```
(kali@kali)-[/home/kali]
PS> sudo apt update
[sudo] contraseña para kali:
Des:1 http://kali.download/kali kali-rolling InRelease [41,5 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 Packages [21,0 MB]
Des:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51,4 MB]
Des:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [120 kB]
Des:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [327 kB]
Des:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [197 kB]
Des:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [911 kB]
Des:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [10,6 kB]
Des:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [26,4 kB]
Descargados 74,0 MB en 19s (3.951 kB/s)
Se pueden actualizar 205 paquetes. Ejecute «apt list --upgradable» para verlos.

(kali@kali)-[/home/kali]
PS> █
```

Se utilizó el comando sudo apt update y luego sudo apt upgrade



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```
Configurando libwireshark18:amd64 (4.4.7-1) ...
Configurando python3-psycopg-c (3.2.6-1) ...
Configurando python3-bottleneck (1.4.2+ds1-2+b1) ...
Configurando python3-numexpr (2.10.2-1+b2) ...
Configurando libgdal36:amd64 (3.10.3+dfsg-1) ...
Configurando kali-linux-core (2025.3.0) ...
Configurando python3-pycaret (4.4.0-2+b3) ...
Configurando initramfs-tools-core (0.148.2) ...
Configurando libldb2:amd64 (2:2.11.0+samba4.22.2+dfsg-1) ...
Configurando kali-system-gui (2025.3.0) ...
Configurando python3-gdal (3.10.3+dfsg-1) ...
Configurando wireshark-common (4.4.7-1) ...
Configurando python3-matplotlib (3.10.1+dfsg1-4) ...
Configurando initramfs-tools (0.148.2) ...
update-initramfs: deferring update (trigger activated)
Configurando wireshark (4.4.7-1) ...
Configurando kali-tools-top10 (2025.3.0) ...
Configurando samba-ls:amd64 (2:4.22.2+dfsg-1) ...
Configurando python3-tables (3.10.2-2) ...
Configurando tshark (4.4.7-1) ...
Configurando samba-common-bin (2:4.22.2+dfsg-1) ...
Configurando samba-dsdb-modules:amd64 (2:4.22.2+dfsg-1) ...
Configurando libsmclient0:amd64 (2:4.22.2+dfsg-1) ...
Configurando samba (2:4.22.2+dfsg-1) ...
smbd.service is a disabled or a static unit not running, not starting it.
nmbd.service is a disabled or a static unit not running, not starting it.
Configurando python3-smbc (1.0.25.1-1+b4) ...
Configurando winbind (2:4.22.2+dfsg-1) ...
winbind.service is a disabled or a static unit not running, not starting it.
Configurando libnss-winbind:amd64 (2:4.22.2+dfsg-1) ...
Configurando smbclient (2:4.22.2+dfsg-1) ...
Configurando libpam-winbind:amd64 (2:4.22.2+dfsg-1) ...
Configurando passing-the-hash (0~2015.12.37+b1) ...
Configurando kali-linux-headless (2025.3.0) ...
Configurando kali-linux-default (2025.3.0) ...
Procesando disparadores para libgl1:amd64 (2.84.1-2) ...
Procesando disparadores para dbus (1.16.2-2) ...
Procesando disparadores para shared-mime-info (2.4-5+b2) ...
Procesando disparadores para debiutils (5.23.1) ...
Procesando disparadores para wordlists (2023.2.0) ...
Procesando disparadores para desktop-file-utils (0.28-1) ...
Procesando disparadores para hicolor-icon-theme (0.18-2) ...
Procesando disparadores para doc-base (0.11.2) ...
Processing 1 changed doc-base file...
Procesando disparadores para libc-bin (2.41-6) ...
Procesando disparadores para man-db (2.13.1-1) ...
Configurando kali-desktop-core (2025.3.0) ...
Procesando disparadores para initramfs-tools (0.148.2) ...
update-initramfs: Generating /boot/initrd.img-6.12.25-amd64
```

3. Configuración del firewall:

3.1 Instalación y habilitación

```
(kali@kali) ~ /home/kali
PS: sudo apt install ufw

Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
crackmapexec libconfig9 libicu-dev libpython3.12-dev libwebp19 python3-nfsclient python3-tomlkit ruby3.1-doc
firebird3.0-common libdirectfb-1.7-7164 libgles-dev libpython3.12-minimal libwext0 python3-ntlm-auth python3-wheel-whl samba-ad-provision
firebird3.0-common-doc libegl-dev libglvnd-core-dev libpython3.12-stdlib libwebp2t python3-packaging-whl python3.12 samba-dsdb-modules
icu-devtools libfcl2t64 libglvnd-dev libpython3.12t64 libx265-289 python3-poetry-dynamic-versioning python3.12-dev python3.12-strongswan
libbabel20230802 libfont2 libgtksourcview-3.0-1 libb2crypt0764 libqt5ensors5 openjdk-23-jre python3-pyinstaller-hooks-contrib python3.12-minimal tdb-tools
libblf1 libfua3 libgtksourcview-3.0-common libb2crypt0764 libqt5webkit5 openjdk-23-jre-headless python3-pyview python3.12-tk
libc++1-19 libgdal35 libgtksourcview-3.0-0v5 libb2crypt19164 libsuperlu6 python3-aioclient python3.12-venv
libc++abi1-19 libgeo2-1.0 libgtksourcview-3.0-0v5 libb2crypt19164 libsuperlu6 python3-aioclient python3.12-venv
libcapstone4 libgl1-mesa-dev libhdf5-103-1t64 libb2crypt19164 libsuperlu6 python3-aioclient python3.12-venv
libconfig9v5 liblapack-dev libhdf5-hl-100t64 libb2crypt19164 libsuperlu6 python3-aioclient python3.12-venv
Utilice «sudo apt autoremove» para eliminarlos.

Installing:
ufw

Paquetes sugeridos:
rsyslog

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 9
Download size: 169 kB
Space needed: 880 kB / 2.918 MB available

Des: http://kali.download/kali-kali-rolling/main amd64 ufw all 0.36.2-9 [169 kB]
Descargados 169 kB en 1s (146 kB/s)
Preconfigurando paquetes ...
Seleccionando el paquete ufw previamente no seleccionado.
(Leyendo la base de datos ... 44102 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../archives/ufw_0.36.2-9_all.deb ...
Desempaquetando ufw (0.36.2-9) ...
Configurando ufw (0.36.2-9) ...
Creating config file /etc/ufw/before.rules with new version
Creating config file /etc/ufw/before.rules with new version
Creating config file /etc/ufw/after.rules with new version
Creating config file /etc/ufw/after.rules with new version
update-rc.d: We have no instructions for the ufw init script.
update-rc.d: It looks like a non-network service, we enable it.
Created symlink /etc/systemd/system/multi-user.target.wants/ufw.service' → '/usr/lib/systemd/system/ufw.service'.
Procesando disparadores para kali-menu (2025.3.0) ...
Procesando disparadores para man-db (2.13.1-1) ...

(kali@kali) ~ /home/kali
PS: █
```

```
Archivo Acciones Editar Vista Ayuda
GNU nano 2.9.3 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none
#PermitRootLogin no
# Logging
#SyslogFacility AUTH
#LogLevel INFO
#PasswordAuthentication no
# Authentication:
PubKeyAuthentication yes
#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubKeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
```

3.2 Cambio de puerto y desactivación de login root

```
(kali@kali)-[/home/kali]
PS> sudo cat /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```



```
# Ciphers and keying
#RekeyLimit default none
PermitRootLogin no
# Logging
#SyslogFacility AUTH
#LogLevel INFO
PasswordAuthentication no
# Authentication:
PubkeyAuthentication yes
#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
```

3.3. Generación de clave mediante RSA.

```
(kali@kalin)-[~]
$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/kali/.ssh/id_rsa):
/home/kali/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase for "/home/kali/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kali/.ssh/id_rsa
Your public key has been saved in /home/kali/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:rH5L6x8W9nPEk1ETrNkiDYKfQTS72bKXWMtkGcS9fq0 kali@kalin
The key's randomart image is:
+--[RSA 4096]--+
|      ++ ... .00|
|      . 0+0 . 0.|
|      ..+.0 *   |
|      . 0+.0B +  |
|      S+0*0 *   .|
|      . .0000 0.|
|      . .00=0 0.|
|      . ..0.. 0E|
|      .0+0.     |
+--[SHA256]--+
```

3.3.1 Copia de la contraseña publica.

```
(kali@kalin)-[~]
$ ssh-copy-id kali@10.0.2.15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: '/home/kali/.ssh/id_rsa.pub'
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: ERROR: ssh: connect to host 10.0.2.15 port 22: Connection refused

(kali@kalin)-[~]
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDQjJ3aG91eF80Kf+Yge7V1qPE1wFTnRgUeZqDFDeckf+Nae3ly0vA3kYdp1E/BsklqeTLYindugTLL53kktwIC3skdV251/AGeFfTegh4Nqub25bqRfAnrX68KCa1cpe17W1P025B5SpZ111g7rk2A+wsNb6a22Nf8ducc39wFJmYBF1Kf3AYWS
53DBIrx0bxc1TWS36H1G077106+wa1rqv38zK3771ge1Djkt0d0Ew1lge+2B1xeuBpPh7qK5w2Av5pmkCACgndowv0x86rdt30cWfDnZAlm201KP21LH1aq+cp3kwl02qer09CEPEQ3kzmqRbXNYlUT5W67QrBWZcSRQ6G81U3JgcCHY+50tSk7zrAmk2e4NhbRg18hQy329uQVCq3KAS
deW9Ar/fATeU7D6/331QWqashJf5ShIHhXm1107YHRFaTGA1v2W8QzK2QeFHRXpuaNbaC65c4cTCgr6QCV3W7y91YB0V1/apPtCwJf4d3B24GwVwQd28TAq6307gM1Bm3yM0qRv53+73ZE/gfyUXEGtSCAHFrV50+tw5bnZmg10urHRC4DR8u8chiwqdy3cmRycCU9+28TFX+Dnop6C1YowGCD
8Nj6eC1Festxp13Q== kali@kalin

(kali@kalin)-[~]
$ mkdir -p ~/.ssh
(kali@kalin)-[~]
$ echo AAAAB3NzaC1yc2EAAAADAQABAAQDQjJ3aG91eF80Kf+Yge7V1qPE1wFTnRgUeZqDFDeckf+Nae3ly0vA3kYdp1E/BsklqeTLYindugTLL53kktwIC3skdV251/AGeFfTegh4Nqub25bqRfAnrX68KCa1cpe17W1P025B5SpZ111g7rk2A+wsNb6a22Nf8ducc39wFJmYBF1Kf3AYWS
53DBIrx0bxc1TWS36H1G077106+wa1rqv38zK3771ge1Djkt0d0Ew1lge+2B1xeuBpPh7qK5w2Av5pmkCACgndowv0x86rdt30cWfDnZAlm201KP21LH1aq+cp3kwl02qer09CEPEQ3kzmqRbXNYlUT5W67QrBWZcSRQ6G81U3JgcCHY+50tSk7zrAmk2e4NhbRg18hQy329uQVCq3KAS
deW9Ar/fATeU7D6/331QWqashJf5ShIHhXm1107YHRFaTGA1v2W8QzK2QeFHRXpuaNbaC65c4cTCgr6QCV3W7y91YB0V1/apPtCwJf4d3B24GwVwQd28TAq6307gM1Bm3yM0qRv53+73ZE/gfyUXEGtSCAHFrV50+tw5bnZmg10urHRC4DR8u8chiwqdy3cmRycCU9+28TFX+Dnop6C1YowGCD
8Nj6eC1Festxp13Q== ~/.ssh/authorized_keys
(kali@kalin)-[~]
$ chmod 600 ~/.ssh/authorized_keys
(kali@kalin)-[~]
$ chmod 700 ~/.ssh
```

4. Configuración de red.

```
(kali㉿kalin)-[~]
$ sudo ufw enable
[sudo] contraseña para kali:
Firewall is active and enabled on system startup

(kali㉿kalin)-[~]
$ sudo ufw allow 2222/tcp
Rule added
Rule added (v6)

(kali㉿kalin)-[~]
$ sudo ufw allow http
Rule added
Rule added (v6)

(kali㉿kalin)-[~]
$ sudo ufw allow https
Rule added
Rule added (v6)

(kali㉿kalin)-[~]
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
2222/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
443 ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
2222/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)
```

Status Configuración del archivo /etc/ssh/sshd_config en Linux



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```
(kali@kali)-[~]
$ cat /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none
PermitRootLogin no
# Logging
#SyslogFacility AUTH
#LogLevel INFO
PasswordAuthentication no
# Authentication:
PubkeyAuthentication yes
#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
```

5. Instalación de auditd.

```
(kali@kali)-[~]
$ sudo apt install auditd
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
crackmapexec libconfig libgles-dev libicu-dev libpython3.12-dev libunwind-10 python3-nflcin python3-tonkit ruby3.1-doc
firebird3.0-common libdirectfb-1.7-764 libgles1 libjxl9 libpython3.12-minimal libutempter0 python3-ntlm-auth python3-wheel-whl samba-ad-provision
firebird3.0-common-doc libgl-dev libglvnd-core-dev liblbfgsb0 libpython3.12-stdlib libwebp-1.3.2 python3-packaging-whl python3-wheel-whl samba-dsdb-modules
icu-devtools libfreetype2 libglvnd-dev libldap-2.5-0 libpython3.12t64 libzstd-1.5.5 python3-poetry-dynamic-versioning python3.12-dev strongswan
libb1-1.0.3 libfontconfig libglvnd-dev libldap2 libpython3.12t64 libzstd-dev python3-pyinstaller-hooks-contrib python3.12-minimal tdb-tools
libb101 libfontconfig1 libglvnd-dev libldap2 libpython3.12t64 libzstd-dev python3-pyview python3.12-tk
libc++abi19 libfontconfig1 libglvnd-dev libldap2 libpython3.12t64 libzstd-dev python3-requests-ntlm python3.12-tk
libcapstone4 libfontconfig1 libglvnd-dev libldap2 libpython3.12t64 libzstd-dev python3-setuptools python3.12-venv ruby-zeitwerk
libconfig9-dev libfontconfig1 libglvnd-dev libldap2 libpython3.12t64 libzstd-dev python3-talloc python3.1-dev
libconfig9-dev libfontconfig1 libglvnd-dev libldap2 libpython3.12t64 libzstd-dev python3-tdb python3.1-dev

Utilice 'sudo apt autoremove' para eliminarlos.

Installing:
auditd

Installing dependencies:
libauparse0t64

Paquetes sugeridos:
auditd-plugins

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 9
Download size: 288 kB
Space needed: 954 kB / 2.907 MB available

Continue? [5/n] s
Des1: http://http.kali.org/kali kali-rolling/main amd64 libauparse0t64 amd64 1:4.0.2-2+b2 (68,6 kB)
Des2: http://http.kali.org/kali kali-rolling/main amd64 auditd amd64 1:4.0.2-2+b2 (217 kB)
Descargados 288 kB en 1s (319 kB/s)
Seleccionando el paquete libauparse0t64:amd64 previamente no seleccionado.
(Leyendo la base de datos ... 444215 ficheros o directorios instalados actualmente.)
Preparando para desempacar ... libauparse0t64:amd64 1:4.0.2-2+b2:amd64.deb ...
Añadiendo 'desviación de /lib/x86_64-linux-gnu/libauparse.so.0 a /lib/x86_64-linux-gnu/libauparse.so.0:usr-is-merged por libauparse0t64'
Añadiendo 'desviación de /lib/x86_64-linux-gnu/libauparse.so.0:usr a /lib/x86_64-linux-gnu/libauparse.so.0:usr-is-merged por libauparse0t64'
Desempaquetando libauparse0t64:amd64 (1:4.0.2-2+b2) ...
Seleccionando el paquete auditd previamente no seleccionado.
Preparando para desempacar ... auditd:amd64 1:4.0.2-2+b2:amd64.deb ...
Desempaquetando auditd (1:4.0.2-2+b2) ...
Configurando libauparse0t64:amd64 (1:4.0.2-2+b2) ...
Configurando auditd (1:4.0.2-2+b2) ...
update-rc.d: We have no instructions for the auditd init script.
update-rc.d: It looks like a non-network service, we enable it.
auditd.service is a disabled or a static unit, not starting it.
auditd.service is a disabled or a static unit, not starting it.
Procesando disparadores para man-db (2.13.3-1) ...
Procesando disparadores para man-db (2025.3.0) ...
Procesando disparadores para libc-bin (2.41-6) ...
```

5.1. Estado de auditd



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```
(kali@kalin)-[~]
$ sudo systemctl start auditd

(kali@kalin)-[~]
$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' -> '/usr/lib/systemd/system/auditd.service'.

(kali@kalin)-[~]
$ sudo systemctl status auditd
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-06-22 13:53:20 -04; 1min 3s ago
  Invocation: d3718c0cd971400181029a63bf877ace
     Docs: man:auditd(8)
          https://github.com/linux-audit/audit-documentation
    Main PID: 222942 (auditd)
      Tasks: 2 (limit: 10499)
     Memory: 624K (peak: 1.8M)
        CPU: 60ms
    CGroup: /system.slice/auditd.service
           └─222942 /usr/sbin/auditd

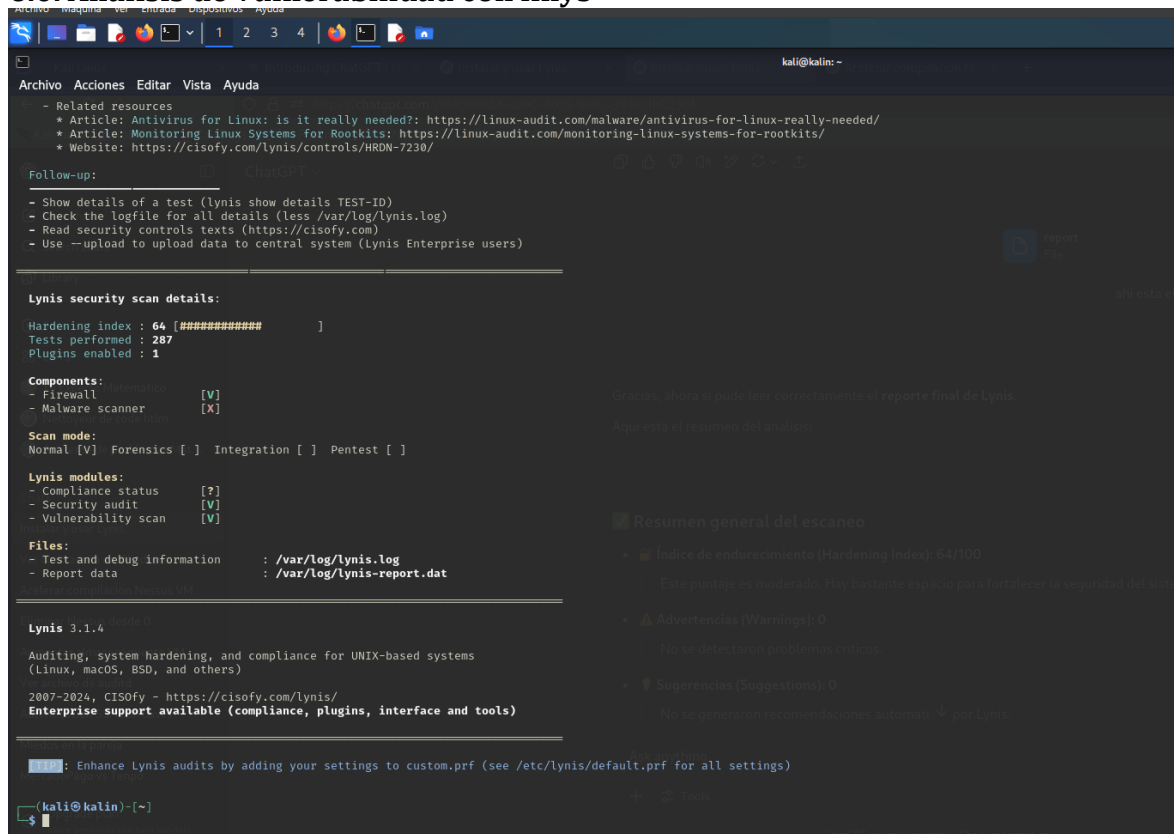
jun 22 13:53:20 kalin systemd[1]: Starting auditd.service - Security Audit Logging Service ...
jun 22 13:53:20 kalin auditd[222942]: No plugins found, not dispatching events
jun 22 13:53:20 kalin auditd[222942]: Init complete, auditd 4.0.2 listening for events (startup state enable)
jun 22 13:53:20 kalin systemd[1]: Started auditd.service - Security Audit Logging Service.
```

5.2. Configuración de auditd.

```
(kali@kalin)-[~]
$ sudo cat /etc/audit/auditd.conf
#
# This file controls the configuration of the audit daemon
#

local_events = yes
write_logs = yes
log_file = /var/log/audit/audit.log
log_group = adm
log_format = ENRICHED
flush = INCREMENTAL_ASYNC
freq = 50
max_log_file = 8
num_logs = 5
priority_boost = 4
name_format = NONE
##name = mydomain
max_log_file_action = ROTATE
space_left = 75
space_left_action = SYSLOG
verify_email = yes
action_mail_acct = root
admin_space_left = 50
admin_space_left_action = SUSPEND
disk_full_action = SUSPEND
disk_error_action = SUSPEND
use_libwrap = yes
##tcp_listen_port = 60
tcp_listen_queue = 5
tcp_max_per_addr = 1
##tcp_client_ports = 1024-65535
tcp_client_max_idle = 0
transport = TCP
krb5_principal = auditd
##krb5_key_file = /etc/audit/audit.key
distribute_network = no
q_depth = 2000
overflow_action = SYSLOG
max_restarts = 10
plugin_dir = /etc/audit/plugins.d
end_of_event_timeout = 2
```

6.0. Análisis de vulnerabilidad con lynis



```
Archivo Máquina Ver Entada Dispositivos Ayuda
kati@kalin: ~
Archivo Acciones Editar Vista Ayuda
- Related resources
  * Article: Antivirus for Linux: is it really needed?: https://linux-audit.com/malware/antivirus-for-linux-really-needed/
  * Article: Monitoring Linux Systems for Rootkits: https://linux-audit.com/monitoring-linux-systems-for-rootkits/
  * Website: https://cisofy.com/lynis/controls/HRDN-7230/
Follow-up:
- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

Lynis security scan details:
Hardening index : 64 [##### ]
Tests performed : 287
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

Lynis 3.1.4
Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)
2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

(kali@kalin)-[~]
$
```

7.0 .Análisis del escaneo:

Con respecto a los resultados generados tras este análisis, se puede concluir que:

- El índice de hardening fue de 64 sobre 100, lo cual para las medidas implementadas es medianamente beneficioso.
- No se detectaron advertencias warnings de vulnerabilidades mayores
- No se encontraron sugerencias del sistema de forma automática de vulnerabilidades medianamente peligrosas.

Medidas de endurecimiento recomendadas:

- Configurar de forma más específicas las alertas de seguridad del firewall. (Unidad 2 Hardening e identificación de vulnerabilidades técnicas Hardening y Seguridad en Sistemas Operativos. s/f).
- Configurar un escaner de malware activo como por ejemplo ClamAV.(CIS Debian Linux 11 Benchmark. s/f)
- Utilizar herramientas para detección de intrusiones como AIDE (CIS Debian Linux 11 Benchmark. s/f)

8.0.Bibliografía.

- (S/f). Ninjaone.com. Recuperado el 24 de junio de 2025, de <https://www.ninjaone.com/es/blog/complete-guide-to-systems-hardening/>.
- *Unidad 2 Hardening e identificación de vulnerabilidades técnicas Hardening y Seguridad en Sistemas Operativos.* (s/f). Umayor.cl. Recuperado el 24 de junio de 2025, de https://campusvirtual.umayor.cl/ultra/courses/_161825_1/outline/file/_5311804_1.
- *CIS Debian Linux 11 Benchmark.* (s/f). Recuperado el 24 de junio de 2025, de http://file:///C:/Users/laburgencia/Downloads/CIS_Debian_Linux_11_Benchmark_v1.0.0_ARCHIVE.pdf.